SynapseFHIR: Executive Submission Brief & System Architecture

Autonomous, auditable clinical triage and referral optimization engine powered by open infrastructure and multi-tier model routing.

1. Executive Summary & Track Target

Attribute	Specification Details
Project Name	SynapseFHIR
Target Tracks	Best Apps and Agents Track | Coding and Agentic Engineering Track
Target Bounties	Grand Prize ($20,000) / Top 3 Placements ($6,000–$10,000) | Best Use of Tavily ($3,000)
Submission Deadline	October 30, 2026 (Devpost)
Core Technologies	Nebius Token Factory, NVIDIA Nemotron Models (Nemotron-3-Nano-30B-A3B & Nemotron-3-Ultra-550b-a55b), Tavily Search API, LangGraph, Pydantic, Streamlit, HL7 FHIR R4

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2. Hackathon Overview & Evaluation Criteria

Challenge Mandate

Build a functional AI system hosted on independent, open infrastructure. Submissions must execute via Nebius Token Factory or Nebius AI Cloud and incorporate at least one NVIDIA open-source model (specifically targeting the Nemotron family).

Judging Criteria Alignment

- **Technological Implementation:** Implements multi-tier model routing via Nebius Token Factory. Low-latency entity extraction is handled by Nemotron-3-Nano-30B-A3B, while deep clinical reasoning is routed to Nemotron-3-Ultra-550b-a55b, achieving a ~70% inference cost reduction over monolithic calls.
- **Design:** Delivers a production-grade product experience via a 3-pillar interactive clinician dashboard featuring live telemetry metrics, direct FHIR provenance citations, and visual protocol checklists.
- **Potential Impact:** Targets the $150B outpatient referral crisis by eliminating referral rejections caused by missing baseline diagnostics.
- **Quality of the Idea:** Replaces hallucination-prone LLM summaries with bidirectional FHIR resource tracing, dynamic web guideline verification via Tavily, and trajectory rollback loops.

Submission Deliverables Checklist

- Public Code Repository (GitHub/GitLab) with an approved open-source license (MIT / Apache 2.0).
- Comprehensive README.md with installation, environment configuration, and architectural workflow.
- Hosted, working public demo URL (Streamlit Cloud or Nebius Serverless Endpoint).
- 3-minute or shorter public YouTube walkthrough demonstrating live execution and telemetry logs.
- Written feedback on Nebius Token Factory and NVIDIA Nemotron model serving.

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3. Problem Statement: The Outpatient Referral Crisis

The contemporary primary care provider shortage and siloed Electronic Health Record (EHR) systems have created severe operational failures across specialist transitions:

- **The 50% Referral Drop-Off:** Over 100 million specialist referral requests are initiated annually in the United States. Due to untracked documentation, absent preliminary workups, and fragmented EHR coordination, only half reach final clinical completion.
- **The Diagnostic Workup Gap:** Up to 30% of scheduled specialist consultations fail to include mandatory preliminary diagnostics (e.g., missing renal imaging prior to nephrology intake, or absent baseline ECGs before cardiology consults). 68% of specialists report

receiving zero clinical background documentation before intake, leading to canceled consults, redundant re-referrals, and delayed patient treatment.

- **Financial Waste:** Unoptimized and unproductive specialist consultation slots cost the U.S. healthcare system $150 billion annually, with direct overhead penalties ranging from $200 to $375 per wasted clinical block.

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4. The SynapseFHIR Solution

SynapseFHIR is an autonomous, auditable clinical triage and referral optimization engine. It automates specialist routing, audits patient records against dynamic medical guidelines, and issues verified pre-requisite diagnostic orders before consult slots are scheduled.

Core Innovations

1. **Tiered Open-Model Routing on Nebius Token Factory:** Employs Nemotron-3-Nano-30B-A3B for high-throughput entity extraction and Tavily query formulation, then hands off context to Nemotron-3-Ultra-550b-a55b for multi-step clinical reasoning and safety contraindication analysis.
2. **Dynamic Guideline Grounding (Tavily Search API):** Bypasses brittle hardcoded rules by dynamically retrieving live, verified society guidelines (KDIGO, ACC, AAN) from peer-reviewed repositories (kdigo.org, acc.org, ncbi.nlm.nih.gov).
3. **Deterministic Bidirectional Citations:** Every single assertion in the synthesized referral brief maps explicitly to underlying FHIR resource IDs (Condition/cond-xxx, Observation/obs-xxx), eliminating unsupported hallucinations.
4. **Trajectory Hashing & Rollback Governance:** Computes a cryptographic SHA-256 state hash of the FHIR bundle. If external EHR systems upload completed diagnostics mid-transaction, the replay engine detects state drift and executes an automated rollback to prevent redundant clinical orders and duplicate patient billing.

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5. End-to-End System Architecture

The pipeline processes clinical data deterministically through six isolated execution phases:

1. **Ingestion & Deterministic Hashing:** Parses standard HL7 FHIR R4 bundles into typed Pydantic models and computes an initial SHA-256 state signature.
2. **Fast Clinical Entity Triage (Nemotron-3-Nano-30B-A3B on Token Factory):** Extracts active conditions, vitals, and labs while generating targeted protocol search queries.
3. **Guideline Retrieval (Tavily Search API):** Retrieves evidence-based clinical intake guidelines filtered by trusted medical domains (KDIGO, ACC, AAN, NCBI).
4. **Deep Clinical Synthesis & Gap Analysis (Nemotron-3-Ultra-550b-a55b on Token Factory):** Evaluates protocol requirements (SATISFIED vs. MISSING), audits drug-lab contraindications (e.g., Lisinopril in CKD 4 with hyperkalemia), and synthesizes a structured referral memo with exact source FHIR citations.
5. **Trajectory Governance & State Replay:** Verifies live bundle state against initial hash. Automatically triggers rollback and compensation to cancel duplicate ServiceRequests if changes occurred mid-transaction.
6. **Three-Pillar Clinician & Admin Telemetry Dashboard:** Presents provenance, guideline checklists, synthesized referral letters, and real-time Token Factory cost/token telemetry.

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6. Technical Specifications & Stack

Component	Technical Selection
Model Serving	Nebius Token Factory (OpenAI-compatible REST API)
Fast Entity Triage Model	nvidia/nemotron-3-nano-30b-a3b (30B total, 3B active MoE)
Clinical Reasoning Model	nvidia/nemotron-3-ultra-550b-a55b (550B total, 55B active MoE, FP4)
Web Grounding	Tavily Python SDK (`search_depth='advanced'`, medical domain filtering)
Orchestration	LangGraph state machine DAG
Validation	Pydantic v2 schemas

Component	Technical Selection
Frontend / Telemetry	Streamlit interactive 3-pillar layout
Configuration	python-dotenv (.env configuration)

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7. Demonstration Scenarios & Interactive UX

- **Scenario 1 — Diagnostic Gap & Pre-Authorization:** Ingests a synthetic patient exhibiting Stage 3b/4 CKD with elevated potassium on Lisinopril. Nemotron-3-Nano extracts clinical entities, Tavily retrieves KDIGO 2024 standards, and Nemotron-3-Ultra flags an omitted Urine ACR and missing Renal Ultrasound, issuing pre-authorized ServiceRequests alongside safety alerts.
- **Scenario 2 — State Drift & Rollback Trigger:** Simulates a live EHR modification where an ultrasound result is uploaded during transaction execution. The TrajectoryGovernor detects the hash mismatch, logs a `ROLLBACK_COMPENSATE` event, and cancels redundant diagnostic orders.

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8. Submission Video Script & Presentation Strategy (3 Minutes)

- **0:00 – 0:30 | The Operational Pain Point:** Cite the 50% referral failure rate and $150B waste driven by missing workups and uncoordinated EHRs. Introduce SynapseFHIR as a verifiable, open-model solution.
- **0:30 – 1:15 | Architecture & Execution Walkthrough:** Trigger Scenario 1 in the UI. Show Nemotron-3-Nano parsing raw FHIR JSON on Nebius Token Factory. Highlight Tavily retrieving live KDIGO intake criteria. Point out Nemotron-3-Ultra isolating the missing Urine ACR test and anchoring every clinical claim to raw FHIR resource IDs.
- **1:15 – 2:00 | Telemetry & Cost Efficiency:** Scroll to the Token Factory Telemetry panel. Explain the multi-tier architectural savings: offloading entity extraction to the 30B MoE model preserves expensive 550B reasoning tokens, cutting inference expenses by ~70% while maintaining deterministic precision.

- **2:00 – 2:40 | Deterministic Rollback Demo:** Select Scenario 2 to simulate external state drift. Demonstrate the automated ROLLBACK compensating action, proving that the agent cannot generate duplicate lab orders if the patient's record updates asynchronously.
- **2:40 – 3:00 | Closing & Compliance:** Conclude on the project's reproducibility: public GitHub repository, permissive MIT license, containerized deployment, and pure open-infrastructure execution via Nebius and NVIDIA.

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9. Final 8-Day Execution Milestones

Timeline	Milestone Focus	Primary Deliverables
Days 1–2	Test Cases	Finalize synthetic FHIR bundles for Cardiology and Endocrinology.
Days 3–4	UX Polish	Add interactive CSS hover highlighting between citations and Pillar 1 FHIR JSON.
Day 5	Deployment	Package FastAPI backend and deploy a public endpoint on Nebius Serverless.
Day 6	Documentation	Polish README.md with token telemetry benchmarks and open-source MIT license.
Day 7	Media	Record, edit, and upload the 3-minute demo video to YouTube.
Day 8	Submission	Complete Devpost fields, submit project, and verify all public URLs.